

COMPREHENSIVE ANALYSIS OF ITSDC, SECUNDERABAD

(For the Visit of Addl CGDA on 06 November 2025)

1.0 INTRODUCTION AND HISTORY

1.1 The Information Technology and Software Development Centre (ITSDC), Secunderabad, was established as part of the *Mission Excel IT* initiative following the cadre restructuring of the Defence Accounts Department (DAD). The establishment was formally approved vide CGDA HQ Letter No. *EDP/177/IT&SDC dated 29 June 2005* addressed to CDA Secunderabad.

1.2 In addition to ITSDC, New Delhi, the CGDA approved the creation of five more ITSDCs at Bangalore, Kolkata, Lucknow, Pune, and Secunderabad, all located within the premises of the respective Regional Training Centres (RTCs).

1.3 The 1st Project developed by IT&SDC is **Project Dolphin**, sanctioned in 2009, was rapidly developed and successfully launched on 01.10.2010. This initiative delivered modern solutions and set a new benchmark for efficiency within its domain. Its swift implementation highlights effective project planning and execution.

1.4 The original authorised strength for ITSDC, Secunderabad, was one JCDA, one DCDA, three SAOs, and six AAOs.

1.5 Vide Authority No. *AN/XII/18505/3/Vol I (PC) dated 04.10.2008 and 19.09.2008*, approval was accorded for the construction of the ITSDC building at Secunderabad. The Chief Construction Engineer (R&D) South, Secunderabad, vide letter No. *22045/SEC/Plg dated 05.01.2012*, confirmed that the building was both physically and financially completed and handed over.

1.6 The Ministry of Defence (Finance), vide ID No. *32(1)/C/2011 dated 26.07.2011*, conveyed the Raksha Mantri's approval for operating a SAG-level post at ITSDC within the overall sanctioned SAG strength of the Department.

1.7 Subsequently, CGDA HQ, vide letter No. *AN/III/3012/Misc/Vol XXVI dated 26.11.2024*, issued Part-I Office Order upgrading ITSDC to SAG level with the following authorised manpower:

(2 IDAS Officers, 3 Senior Accounts Officers (SAO), 6 Assistant Accounts Officers (AAO), 10 Auditors, 2 Stenographers, 3 MTS. **Total Authorised Strength: 26 Personnel**)

2.0 PROJECTS IN HAND

ITSDC, Secunderabad, is currently managing nine major software systems, serving different functional domains of the Defence Accounts Department. A summary of the key projects is provided below.

Project	Start Year	No of Tables	No of Foreign Keys	DB Size Biggest	Lines Of code	Total Screen	Api	SAMANVYAY Tickets (Total)
Dolphin	2009	208	933	145 GB	484142	563	2	807
Skylark	2012	Handed Over to PAO GREF Pune in 2017						
Tulip 1 & PMS	2014	350	2645	35 GB	1490689	2097	11	1443
Dak Ghar	2014	11	37	23 MB	19337	10	0	0
IRIS	2017	142	797	5 GB	350000	600	0	0
Honey B	2019	37	30	9 GB	77924	128	04	0
Tulip 2.0	2024	365	2901	30 GB	589526	593	6	0
Falcon	2024	186	829	30 GB	469929	514	2	186
SAMANVYAY	2024	21	30	5 MB	40000	40	0	11
Phoenix	2025	83	232	6.6 GB	275715	195	0	195

3.0 PROJECTS IN PIPELINE

The following projects are under planning, design, or developmental stages:

Project	Current Status
Tulip PMS 2.0	Version 1.0 is currently running. Tech stack Upgradation to be done
IRIS 2.0	Version 1.0 is currently running. Tech stack Upgradation to be done
Honey B 2.0	Version 1.0 is currently running. Tech stack Upgradation to be done
PSU Module	Tables + Models + Controllers
MoD Pay Module	Rolled out. Tech stack Upgradation to be done

Foreign Pay Module	Development Finished, Integration and testing Pending. Tech Stack Upgradation to be done.
Gyan Saathi AI Chatbot	Development finished. AI Server also Purchased. Testing in progress
AI Forensic Audit	Pilot version already rolled out. Expansion planned
AI Translation	Developed, Planned to integrate with Gyan Saathi.
Data Analysis	Architecture created. Presented in SoM. Expansion planned.

4.0 MANPOWER DETAILS

4.1 : Manpower in ITSDC (Designation Wise)

Category	Authorised	Posted
IDAS	2	3
SAO	3	9
AAO	6	6
Auditor	10	12
Steno	2	1
MTS	3	1 (Attached)
TOTAL	26 Nos	32 Nos

Outsourced IT Contract Manpower:- 5 Nos

Conservancy Manpower :- 4 Nos

4.2 : Manpower in ITSDC (Skill Wise)

Skill	Posted
Administration (IDAS)	3
Developer	21
System Admin	1

Cyber Specialist	3
Document Writer	1
Maintainer / Operator	1
STENO	1
MTS	1
TOTAL	32 Nos

4.3 Manpower Scaled Comparative Analysis: IT&SDC vs. The Similar App. Developers

4.3.1: Comparison with Private Players in the market

The table below contrasts our current manpower against the estimated staffing that **Oracle (NetSuite/Fusion ERP)**, **SAP**, and **Workday** would allocate for a comparable 15-project portfolio. The benchmarks are derived from their known team structures and standard industry ratios.

Role	IT&SDC Working Strength (for 15 Projects)	Oracle (NetSuite/Fusion ERP) (Est. for 15 Projects)	SAP (Est. for 15 Projects)	Workday (Est. for 15 Projects)	The Critical Manpower GAP (vs. Average)
Core Developers	21	90 (Based on ~6 developers per project team)	90 (Similar enterprise project team size)	90 (Similar enterprise project team size)	-69
Quality Assurance (QA) / Testers	1	30-45 (Dedicated QA specialists per implementation)	30-45 (1-2 dedicated testers per project team)	30-45 (Similar enterprise standard)	~ -40
DevOps / System Administrators	1	15 (Dedicated Service/Security Admins per project)	15 (Dedicated roles for deployment)	15 (Specialized roles for system administration)	-14

			and infrastructure)		
User Interface (UI) Developers	3	15 (Dedicated UI/UX roles for usability)	15 (Dedicated designers for user experience)	15 (Focus on consistent user interface across modules)	-12
AI / Data Scientists	0	30 (Dedicated teams for AI/ML in Fusion ERP)	30 (Specialized teams for ML and analytics functions)	30 (Dedicated teams for AI/ML features in financials)	-30
Documentation / Business Analysts	1	30 (Dedicated Business Analysts per project)	30 (2 Analysts per team)	30 (Analysts to translate business needs)	-29

4.3.2: Comparison with other Government accounting and payment packages

Role	IT&SDC Sanctioned Strength (for 15 Projects)	PFMS Model (Estimated for a comparable portfolio)	State IFMS / Treasury Systems (Estimated for a comparable portfolio)	The Critical Manpower GAP (vs. GOI Standard)
Core Developers	21	~75-90 (Based on multiple RFPs for outsourced Sr./Jr. Developers)	~75-90 (Similar tender-based staffing)	-50 to -69
Quality Assurance (QA) / Testers	1	~25-30 (Dedicated QA and testing teams are standard in SI contracts)	~25-30 (Mandated in development RFPs)	~ -27
DevOps / System Administrators	1	~15-20 (Multiple tenders for Sr. System Admins, DBAs, etc.)	~15-20 (Dedicated roles for server and DB management)	~ -16
User Interface (UI) Developers	3	~15 (Specialized roles within outsourced teams)	~15 (Modernization projects require UI specialists)	-12
AI / Data Scientists	0	~10-20 (New capabilities sourced via specialized contracts)	~10-20 (Emerging requirement for analytics)	-10 to -20

Documentation / Business Analysts	1	~15-20 (Consultants and analysts are key roles in tenders)	~15-20 (Essential for requirement gathering and documentation)	~ -16
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5.0 Budget Allocation of ITSDC for the year 2025-26

Category / Code Head	Sanctioned Amount
ICT (045/08)	96,50,000
TA/DA (094/30)	18,00,000
Office Expense (094/30)	25,00,000
Rent/ Others (094/37)	6,00,000
Repair & Maintenance (094/40)	1,25,000
Other Revenue (094/41)	1,80,000
Digital Equipment (094/94)	1,00,000
TOTAL	1.49 Crores

6.0 : IT INVENTORY

CATEGORY	Available
Server (AI)	01
Desktop	38
Laptop	32
AI Laptop	03
Printers	09
Firewall	01
NAS Device	01
Interactive Panel	1

7.0 TECHNOLOGY STACK

All ITSDC projects are built using a robust open-source stack comprising: React JS, Spring Boot, Java, Apache, and PostgreSQL with Linux OS

This ensures scalability, interoperability, and long-term sustainability for departmental applications.

8.0 PROJECT BRIEFS

8.1 DOLPHIN (JCOs/ORs Pay and Allowances System)

DOLPHIN replaces legacy COBOL-based systems with a modern digital platform for JCOs/ORs pay management. It caters to real-time receipt, audit, processing of DO II, and monthly payments, AFPP fund management, and final settlements with built-in automation and Appendix J compliance for approximately 13 lakh Army personnel (JCOs/ORs), managing over ₹2.20 lakh crore annually.

Pay fixation for the last three pay commissions (5th, 6th, and 7th) is handled in Dolphin, using legacy data from 1998. Operating within a secure, closed DAD network, DOLPHIN enhances accuracy, reduces rejections, and ensures audit readiness through MIS dashboards and structured approval workflows.

8.2 PHOENIX (Agniveer Pay System)

PHOENIX is an enterprise-grade pay system developed for Agniveers under the Agnipath scheme and is currently deployed at CAPAO, Delhi Cantt. It manages an annual budget of over ₹9,000 crore for 1.5 lakh personnel, ensuring accuracy and complete policy compliance.

Built on modern frameworks such as Java 17, Spring Boot 3.3, React JS, and PostgreSQL, the platform has reduced DO II processing time from five days to just one day. The earlier Huno system incurred an annual licensing cost of nearly ₹10 crore.

8.3 FALCON (Army Officers Pay System)

FALCON handles pay and entitlements for approximately 60,000 officers, amounting to ₹16,000 crore annually. It automates real-time DO II processing, complex pay fixation, arrears, and deductions in alignment with 7th CPC norms, ensuring transparency and compliance.

The system integrates with DTS and NCS for seamless reconciliation, thereby strengthening data integrity and governance

8.4 TULIP

Tulip is an office automation system designed for regional and functional controllers. It covers the following functions: Third Party Payments, Unit Payments, MRO Adjustments, Budget Monitoring, Civilian Pay System, DAD Pay Bill. Payments worth ₹2.75 lakh crore were processed through Tulip in the year 2024–25.

TULIP includes workflow tracking, dashboards, and integration with e-Rakshaawas, Nidhi 2.0, HoneyB, CBM and NCS for real-time monitoring. It has significantly improved payment timelines, accountability, and audit transparency at regional offices.

9.0 ARTIFICIAL INTELLIGENCE & DATA ANALYSIS INITIATIVES

9.1 AI-Driven Forensic Audit System

This system automates end-to-end financial audits using locally deployed Large Language Models (LLMs) such as Qwen-33B through Ollama. It generates SQL-based forensic queries, performs compliance checks, and produces verifiable PDF reports with evidence.

Developed with Flask, SQLAlchemy, Pandas, and PostgreSQL, it ensures complete data sovereignty and significantly reduces audit turnaround time.

9.2 AI Translation Tool for Bilingual Part-II Orders

This tool automates translation of departmental Part-II Orders from English to Hindi using OCR and the mBART-50 multilingual model. It processes scanned or typed documents, offering real-time previews and bilingual PDF outputs.

The system supports compliance with official language requirements and drastically reduces manual translation effort.

9.3 GYAN SATHI (AI Chatbot)

GYAN SATHI is a Retrieval-Augmented Generation (RAG) chatbot powered by LLaMA 3.3 (70B). It is designed to respond to procedural and audit-related queries using more than 50 official manuals. Hosted securely on DAD infrastructure, it delivers responses with 95% accuracy and enhances decision-making speed by over 80%.

Planned features include Hindi translation and real-time chat capabilities for drafting office notes, official letters, and other forms of office correspondence.

9.4 National Level Data Integration & Analytics Platform

NLDIAP functions as a centralized data warehouse integrating information from DOLPHIN, PHOENIX, and SIFA. It facilitates cross-domain analytics, predictive insights, and policy formulation through visual dashboards.

10.0 CYBER SECURITY INITIATIVES

1. Conduct of voluntary cybersecurity lectures and awareness programs.
2. Continuous dark web monitoring and threat intelligence gathering
3. Vulnerability and Malware Analysis
4. Security Operations Centre (SOC) monitoring
5. Recent data leaks identified
 - a. Malicious Dolphin Circular
 - b. Malicious phishing Domains mimicking NIC mail Login
 - c. CGDA website leak analysis
 - d. Flagged 'purported' Kavach App breach.
 - e. Leaked departmental Email, PCDA (O) Pune website login credentials flagged.

11.0 Submissions with CGDA HQ

- a) SAAMARTH System Integration
 - ERP developed by the Indian Army to replace existing HRMS.
 - Will serve as a data source for DOLPHIN and FALCON.
 - Pilot testing completed at PAO RRRC Delhi.
 - Rejection rate currently above 5%, requiring optimization.

b) Manpower Contract

Proposal under submission to Addl CGDA for approval.

c) Staff Car

Returned by CGDA HQ; requisition for replacement under consideration.

12.0 CONCLUSION

ITSDC, Secunderabad, has become a core technology centre of the Defence Accounts Department, leading major digital transformation initiatives. It has successfully developed and maintained key systems like DOLPHIN, PHOENIX, FALCON, and TULIP, enhancing efficiency and transparency. AI and data-driven tools such as Forensic Audit, Gyan Saathi, and Translation Engine reflect its growing technical capability.

Despite limited manpower, the Centre has delivered enterprise-level solutions comparable to national platforms. Its open-source technology stack ensures scalability, interoperability, and security. The team's expertise supports end-to-end automation across pay, audit, and accounting functions. Cybersecurity, AI, and analytics initiatives further strengthen DAD's digital ecosystem. Manpower augmentation and infrastructure support from CGDA HQ remain crucial for sustained progress.

ITSDC is poised to evolve into a “**Centre of Excellence**” for AI, analytics, and software innovation. Its work underlines DAD's commitment to a transparent, efficient, and technology-driven future.